Project Presentation

A Hybrid NLP Framework for Automated Scoring of Privacy Policy for mHealth apps

BY

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Introduction

- Privacy policies are the primary means for companies to communicate data practices
- These documents are often complex, technical, and difficult to evaluate
- Manual evaluation is time-consuming, subjective, and inconsistent
- Users rarely read or comprehend these critical documents

Objectives

Primary Goal: To develop and validate an automated system for evaluating privacy policy quality across multiple dimensions

Specific Objectives:

- Create Enhanced Feature Extraction:Develop comprehensive terminology lists for each policy dimension
- Identify key indicators of quality in privacy policies
- Build regression based machine learning models to predict quality ratings for 11 dimensions
- Validate on real-world privacy policies
- Analyze Dimension Predictability:Determine which policy dimensions can be most reliably evaluated
- Identify features that best predict policy quality
- Provide actionable insights for privacy policy improvement

Methodology

DATA PREPROCESSING

Data Collection:

- Selected 30 privacy policies from mobile health applications
- Split into 25 training examples and 5 test examples
- Policies represent diverse quality levels and writing styles

Evaluation Using Multi-Dimensional Rubric:

- Each policy rated on 11 dimensions using a 5-point scale (1-5)
- Structured evaluation rubric with clear criteria for each dimension

Dataset Characteristics:

- Training set: 25 policies × 11 dimensions = 275 ratings
- Test set: 4 policies × 11 dimensions = 44 ratings
- Text length range: 3000+ words per policy
- Quality distribution ensures model exposure to full rating range

Example Rubric:

Rating	Transparency (Grade Level)	Security (Detail)	Regulatory Compliance
5 (Excellent)	Simple language, 8th grade reading level or below	Detailed encryption & security protocols explained	Comprehensive coverage of all applicable regulations
4 (Good)	Generally accessible, some complex sections	Mentions specific security measures	Covers major regulations with some details
3 (Adequate)	Mix of simple and complex language	Basic security statements provided	Mentions primary regulatory frameworks
2 (Poor)	Highly technical, difficult reading	Vague security assurances	Minimal regulatory references
1 (Very Poor)	Legal jargon, college-level reading	No security information	No regulatory compliance info

Methodology

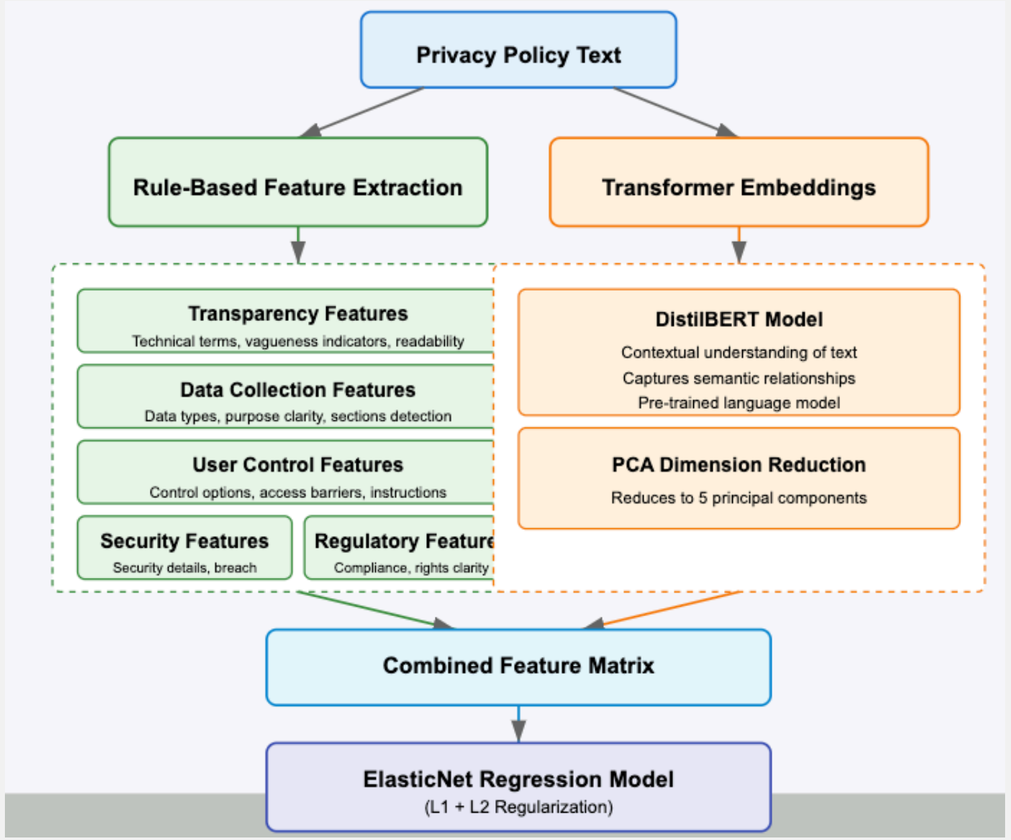
MODEL ARCHITECTURE

Rule-Based Feature Extraction:

- Custom-built extractor with expanded terminology lists
- 5 feature categories aligned with quality dimensions:
  - Transparency: Technical terms, vagueness indicators, readability metrics
  - Data Collection: Data types mentioned, purpose clarity terms
  - User Control: Control options, access barriers, instructions
  - Security: Security details, breach notifications, encryption
  - Regulatory: Regulations mentioned, rights clarity, legal basis

Transformer Embeddings:

  - DistilBERT model to capture semantic meaning
  - PCA dimension reduction to prevent overfitting
  - 5 principal components capturing core semantic patterns



Methodology

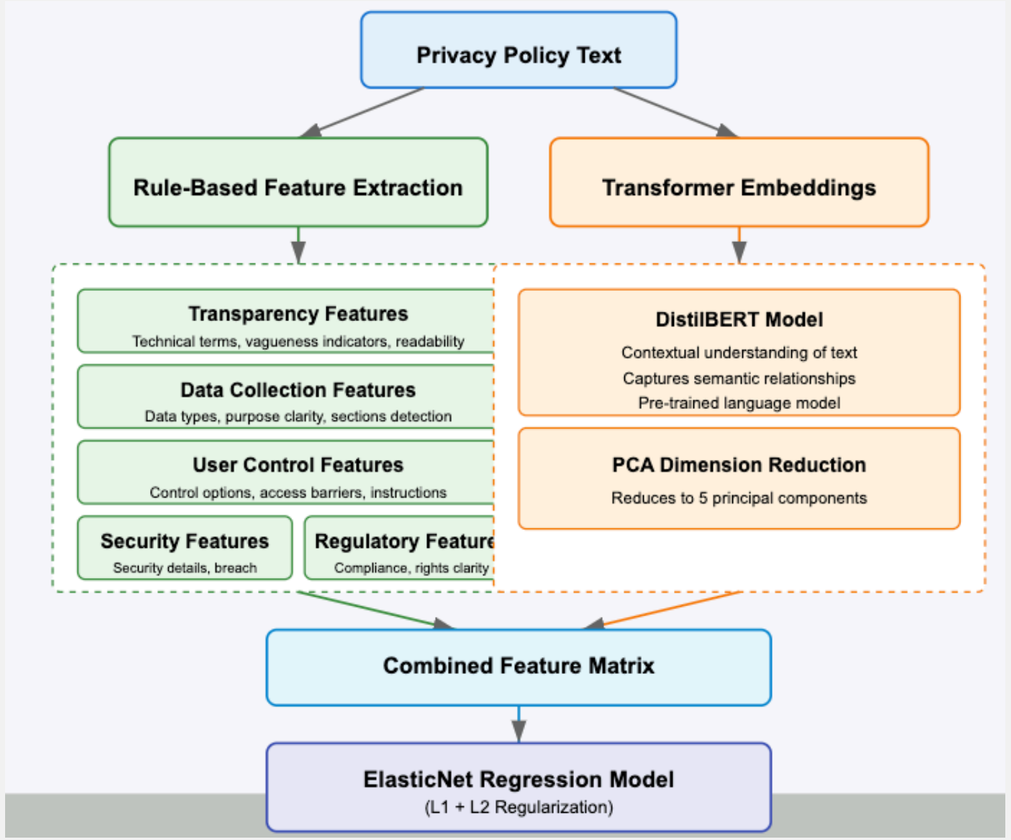
MODEL ARCHITECTURE & TRAINING

Model Selection:

- ElasticNet Regression with combined L1 and L2 regularization
- Alpha = 0.5, L1 ratio = 0.7 for effective feature selection
- Standardized features for consistent scale

Training Process:

- 25 training examples × 11 dimensions
- Feature matrix with 69 features per policy
- Independent model training for each dimension
- Test evaluation on 4 unseen policies



Results

PERFORMANCE BY CATEGORY

Security (best performing):

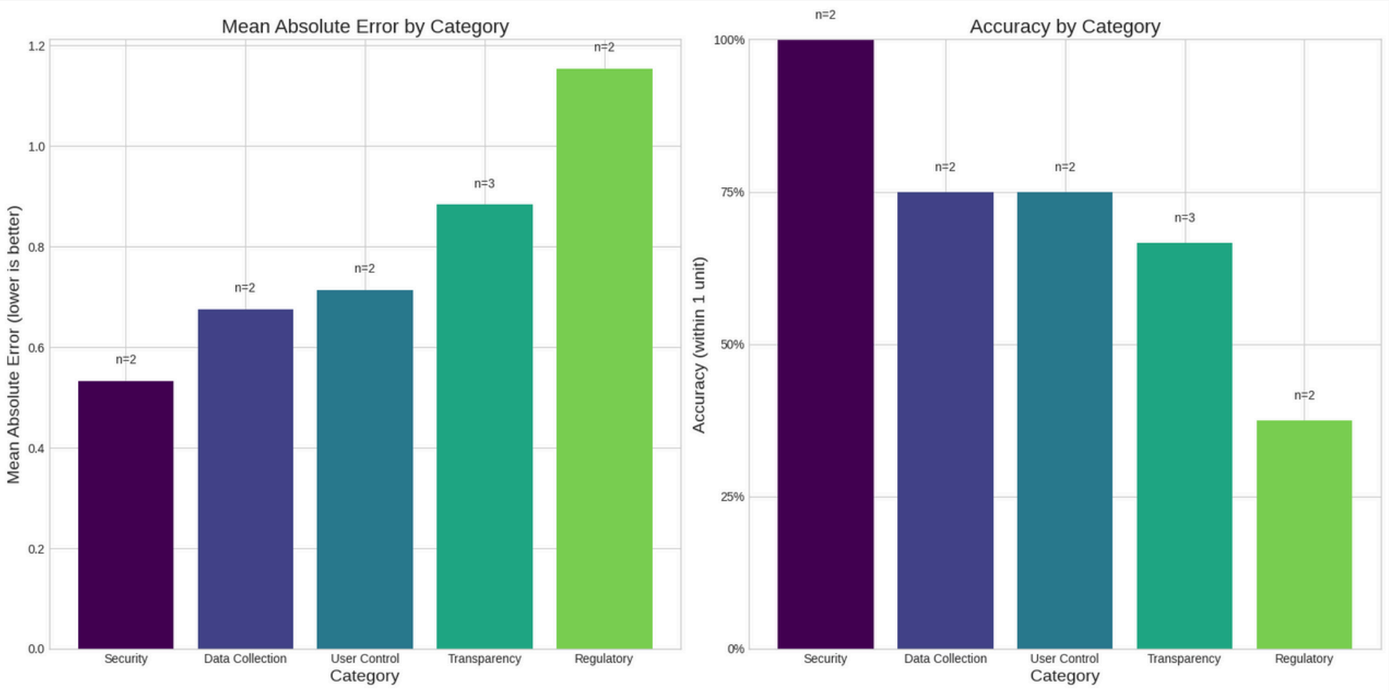
- MAE: 0.53 (lowest error)
- Accuracy: 100% (all predictions within 1 unit)
- High predictability due to specific, standardized security terminology
- 2 dimensions: security_detail and breach_notification

Data Collection:

- MAE: 0.67 (second best)
- Accuracy: 75%
- Consistent pattern detection in data type descriptions
- 2 dimensions: specificity and purpose_clarity

User Control:

- MAE: 0.71
- Accuracy: 75%
- Reliable identification of control options and barriers
- 2 dimensions: control_options and access_barrier



Results

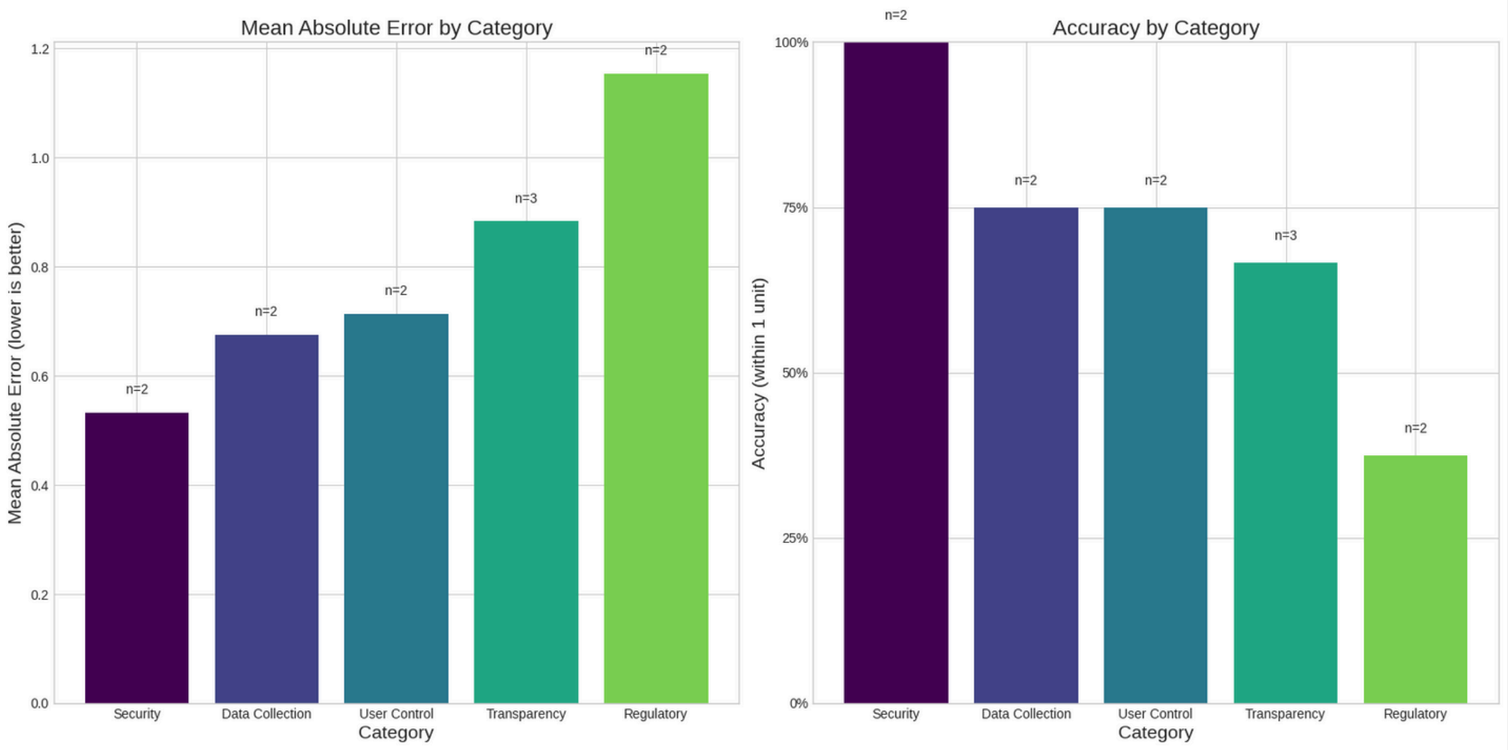
PERFORMANCE BY CATEGORY

Transparency:

- MAE: 0.89
- Accuracy: 67%
- More challenging due to subjective nature
- 3 dimensions: grade_level, term_density, vagueness_index

Regulatory Compliance (most challenging):

- MAE: 1.15 (highest error)
- Accuracy: 38% (lowest)
- Complex legal language with high variability
- 2 dimensions: regulatory_coverage and rights_clarity



Results

DIMENSION-LEVEL PERFORMANCE

Top Performing Dimensions:

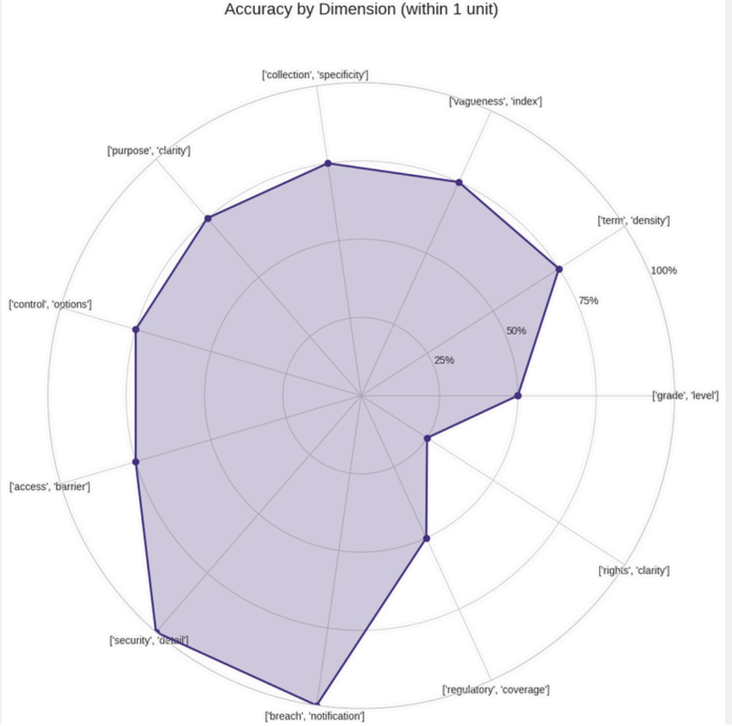
- security_security_detail: MAE 0.41, 100% accuracy
- security_breach_notification: MAE 0.65, 100% accuracy
- user_control_access_barrier: MAE 0.57, 75% accuracy
- data_collection_purpose_clarity: MAE 0.57, 75% accuracy

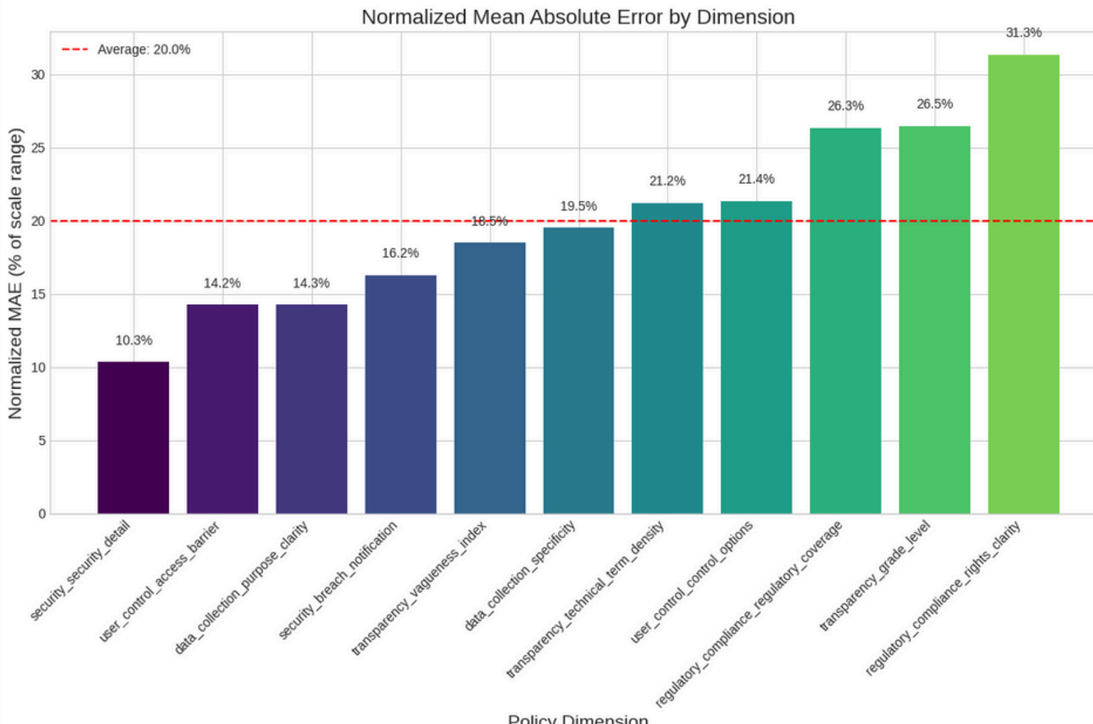
Most Challenging Dimensions:

- regulatory_compliance_rights_clarity: MAE 1.25, 25% accuracy
- transparency_grade_level: MAE 1.06, 50% accuracy
- regulatory_compliance_regulatory_coverage: MAE 1.05, 50% accuracy

Insights:

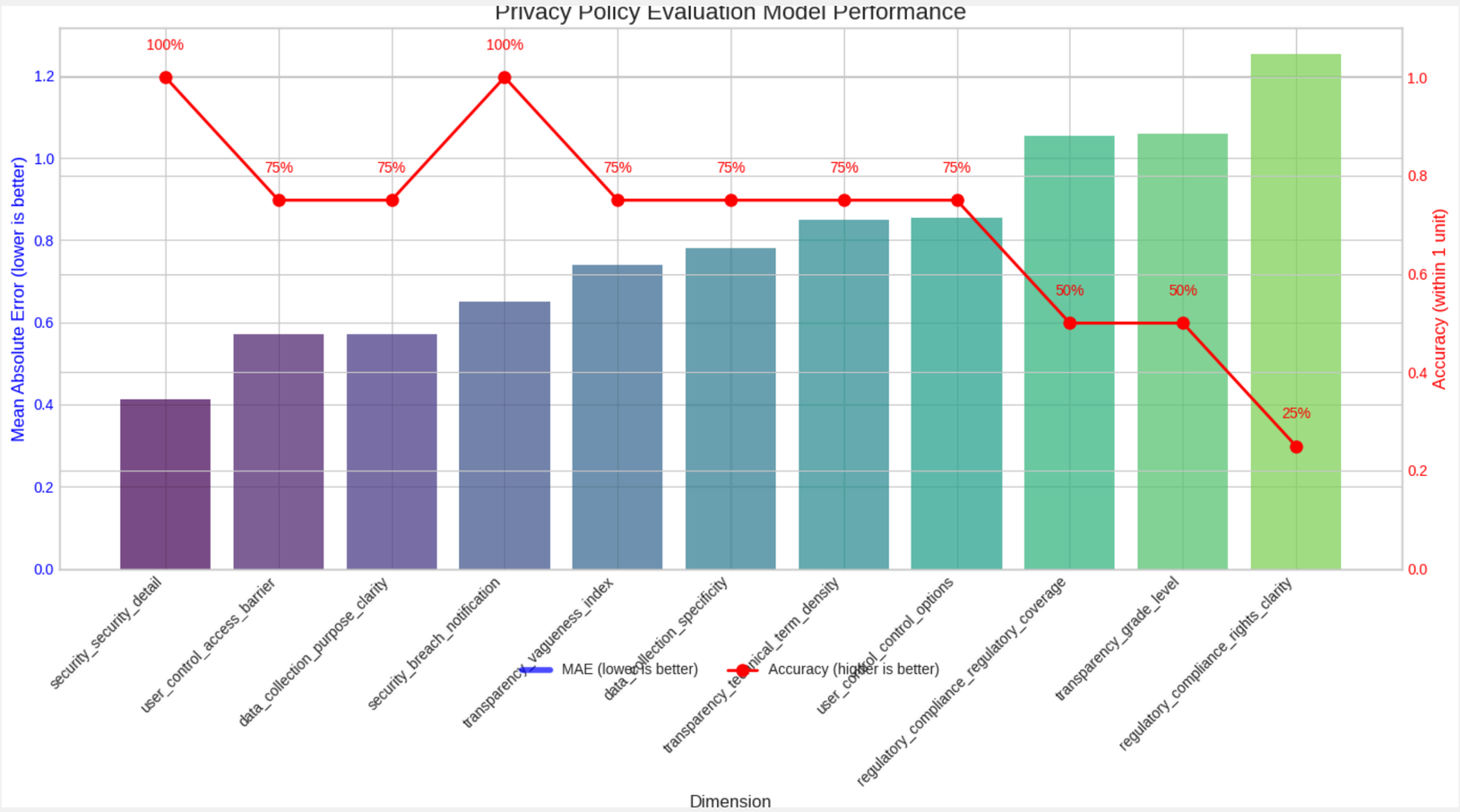
- Security dimensions show strongest predictability (10.3% normalized MAE)
- Regulatory dimensions need further refinement (> 26% normalized MAE)
- Overall model accuracy: 70.5% within 1 unit
- Average normalized MAE: 20.0% of scale range
- Performance pattern suggests training on more examples would enhance results





Results

DIMENSION-LEVEL PERFORMANCE



Results

HOW OUR MODEL CAPTURES IMPORTANT POLICY ELEMENTS

Key Patterns Discovered:

- Legal & Technical Terms Matter: Features detecting specific legal basis mentions strongly predict user control quality
- Direct Term Mentions Drive Predictions: Explicit mentions of encryption are key predictors for security scores
- Purpose Clarity Crosses Dimensions: Purpose clarity features predict both data collection and regulatory compliance quality
- Transformer Embeddings Capture Nuance: Embedding features (emb_2) contribute significantly to access barrier prediction

Zeros in Feature Importance:

- Model performs feature selection through L1 regularization
- Many basic features (word_count, sentence_count) eliminated in favor of specific term matches
- Demonstrates model is identifying targeted indicators rather than general text statistics

Dimension-Specific Prediction Mechanisms:

- Security dimensions: Relies on explicit security terminology
- User control: Identifies legal basis mentions as quality indicators
- Regulatory compliance: Leverages purpose clarity as a proxy for quality

Future Work

Expand Dataset Size

- Increase training samples beyond 25 policies for better generalization
- Add more test examples to validate model robustness

Advanced Model Architectures

- Test ensemble methods combining multiple model types
- Explore fine-tuned language models specific to legal/privacy text
- Implement attention mechanisms to capture complex relationships
- Investigate zero-shot and few-shot learning for new dimensions

Practical Applications

- Policy Assessment Tool for automatic quality evaluation
- Interactive Dashboard for privacy professionals to analyze policies
- Recommendation System suggesting specific improvements

Cross-Policy Comparison

- Automated benchmarking against industry standards
- Competitive analysis between organizations in the same sector
- Historical tracking of policy changes and improvements

Thank you